

Mohit Mahey

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Education

Ph.D. Crop and Soil Sciences, Candidate

August 2026

Michigan State University, East Lansing, U.S.A

B.Sc. Agriculture (Honors)

July 2021

Punjab Agricultural University, Ludhiana, India

Research Experience

Michigan State University, East Lansing, Michigan, USA

Post-Doctoral Researcher

May 2026 - current

with Dr. Timothy Miles

- Developmental Expression atlas of the resistant and susceptible isolates of fungi *Erysiphe nicotri* Using spatial transcriptomics
- Transcriptomics analysis to understand Fungicide resistance in *Erysiphe nicotri* to quinoxifen
- Population diversity and structure of oak wilt fungi, *Bretziella fagacearum* across USA

Graduate Research Assistant

August 2021 - May 2026

with Dr. Eric L. Patterson and Dr. Peter K. Lundquist

- Investigating site of action of herbicide indaziflam through use of cytology, biochemistry and transcriptomics
- Understanding resistance mechanisms in *Poa annua* towards multiple chemistries
- Evaluating the effects of indaziflam outside of plant kingdom
- Variant calling from transcriptomics data to find variants involved in resistance
- Application of malathion on *Poa annua* causes reversal of resistance towards HPPD and ALS inhibitors
- Population structure of *Amaranthus palmeri* invasion of the Mediterranean
- Orthogonal approach in finding site of action of indaziflam using in-silico molecular docking studies.
- Using molecular docking to find potential cytochrome p450s responsible for herbicide resistance
- Population structure of *Erythroxylum coca* across Colombia

Georg-August University of Göttingen, Germany

Intern/Visiting scholar

February 2025 - May 2025

with Dr. Rebecka Dcker

- Reclassification of HRAC group 15 herbicides
- Transformation of KCSs genes in yeast (*Saccharomyces cerevisiae*) and herbicide feeding assays
- Extraction of fatty acids and profiling using GC-MS

Punjab Agricultural University, Ludhiana, Punjab, India

Undergraduate Research Assistant (School of Biotechnology)

Dec 2020 - March 2021

with Dr. Dharminder Bhatia

- Application of CRISPR/Cas9 to insert the *osmtl* gene in rice to produce rice haploid plants
- Use of PCR, polyacrylamide, and agarose gel electrophoresis
- Nanodrop quantification to test concentration of extracted DNA from rice
- In-vivo agrobacterium transformation of rice embryos using gene gun
- Culturing transformed plants by induction of callus

with Dr. Manmeet Kaur

- Organized multiple camps in various villages to inform farmers about harmful effects of paddy-straw burning
- Visited multiple farmers' fields for soil sample collection and pest surveys
- Demonstration of alternatives to paddy-straw burning such as Super Seeder machine, Happy Seeder machine, Mould Board plough, and Baler to farmers
- Application of Integrated Pest Management on farmer fields as a pilot model to show its benefits
- Surveyed village Batura for agricultural practices like crop rotation and irrigation sources to get a better understanding of actual practices
- Conducted survey of hundreds of farmers in Jalandhar district regarding use of rice transplanter, direct seeded rice technology, and yield test of new varieties of rice, PR128 and PR129
- Plant care for citrus, guava trees, onion, garlic, cauliflower and tomatoes
- Evaluating urban gardens for cost effectiveness, ease of use, and practicality of a soil-less method of growing vegetables
- Made an insect specimens box and drip-irrigation model for farmers' training, layout, preparation, and sowing of experimental plots to test various varieties of wheat

Practical Crop Production Trainee (Department of Agronomy)

May 2019 - March 2020

with Dr. Kulvir Singh Saini

- Responsible for producing a crop of *Cajanus cajan* and wheat in rotation on 1 acre from seeding to marketing with special attention to pest management

Publications

A Bioassay to Determine *Poa Annua* Responses to Indaziflam

Pritchard, Benjamin D., Jose J. Vargas, Mohit Mahey, James T. Brosnan, and Eric L. Patterson - *Frontiers in Agronomy* 5 (April 2023)

Cyanoglobule lipid droplet accumulation as a stress response to nitrogen starvation in a non-N₂-fixing mutant strain of *Anabaena* sp. PCC 7120

Febri A. Susanto, Carrie Hiser, Mohit Mahey, Peter K. Lundquist - *PloS one*, February 20, 2026, e0343220

Dynamic Changes to the Plastoglobule Lipidome and Proteome in Maize over a Dehydration/Rehydration Cycle

Devadasu, Elsinraju, Anthony L Schillmiller, Mohit Mahey, Natali Gonzalez, Febri A Susanto, and Peter K Lundquist - *Journal of Experimental Botany*, October 14, 2025, eraf453

Dynamic Changes to the Plastoglobule Lipidome and Proteome in Maize during Heat Stress and Recovery

Devadasu, Elsinraju, Febri A Susanto, Anthony L Schillmiller, Mohit Mahey, Cassandra Johnny, and Peter K Lundquist - *Journal of Experimental Botany*, October 14, 2025, eraf452

A High-Quality Genome Resource for the Oak Wilt Pathogen *Bretziella Fagacearum*

Chahal, Karandeep, Mohit Mahey, Carmen Medina-Mora, et al. - *PhytoFrontiers* 5, no. 1 (2025): 11216

Presentations

Cytochrome P450s mediated multiple herbicide resistance in *Poa annua* (July 2025)

Mohit Mahey, Jinyi Chen, Michael Ozolins, Peter K. Lundquist, Eric L. Patterson - *Plant Biology*, 2025: Poster Presentation

A combined transcriptomic and metabolomic approach to understanding the impact of drought stress on glyphosate resistance in Horseweed (*Erigeron canadensis* L.) (February 2025)

Michael Ozolins, Mohit Mahey, Sara Ivarez-Rodriguez, Ahmet Tansel Serim, Eric Patterson - *Weed Science Society of America*, 2025: Oral Presentation

Applications of Next Generation Protein Folding and Docking Simulations to Study Herbicide Mode-of-Action (February 2025)

Sara Ivarez-Rodriguez, Mohit Mahey, Eric L. Patterson - Weed Science Society of America, 2025: Poster Presentation

Multiple herbicide resistance in *Poa annua* is mediated by Cytochrome P450s (December 2024)

Mohit Mahey, Peter K. Lundquist, Eric L. Patterson - North Central Weed Science Society, 2024: Oral Presentation

Understanding interaction of cytochrome p450 with herbicides in *Poa annua* (October 2024)

Mohit Mahey, Peter K. Lundquist, Eric L. Patterson - Artificial intelligence in Drug Discovery, 2024: Poster Presentation

Dissecting metabolic resistance in *Poa annua* (June 2024)

Mohit Mahey, Jinyi Chen, Peter K. Lundquist, Eric L. Patterson - Plant Biology, 2024: Poster Presentation

Transcriptome Analysis to Identify Genes Involved in Metabolic Resistance to Multiple Herbicides in Palmer amaranth (*Amaranthus palmeri*) (January 2024)

Rishabh Singh, Yaiphabi Kumam, Mohit Mahey, Eric Patterson, Sanzhen Liu, Mithila Jugulam -

Gene co-expression network analysis in indaziflam resistant and susceptible *Poa annua* (December 2023)

Mohit Mahey, Jinyi Chen, Peter K. Lundquist, Eric L. Patterson - North Central Weed Science Society, 2023: Poster Presentation

Gene co-expression network analysis in indaziflam resistant and susceptible *Poa annua* (August 2023)

Mohit Mahey, Jinyi Chen, Peter K. Lundquist, Eric L. Patterson - Plant Biology, 2023: Poster Presentation

Discovering site of action of herbicide Indaziflam (January 2023)

Mohit Mahey, Jinyi Chen, Peter K. Lundquist, Eric L. Patterson - International Weed Genomics Consortium, 2023: Poster Presentation

Discovering site of action of herbicide Indaziflam (December 2022)

Mohit Mahey, Jinyi Chen, Peter K. Lundquist, Eric L. Patterson - North Central Weed Science Society, 2022: Poster Presentation

Discovering site of action of herbicide Indaziflam (July 2022)

Mohit Mahey, Jinyi Chen, Peter K. Lundquist, Eric L. Patterson - Plant Biology, 2022: Poster Presentation

Teaching

Teaching Assistant, Biochemistry and Molecular Biology, MSU

January 2026 - May 2026

Introductory Biochemistry Laboratory

- Setting up labs and helping students with experiments
- Grading the lab notebooks
- Helping the instructor with day-to-day tasks

Teaching Assistant, Biochemistry and Molecular Biology, MSU

August 2025 - December 2025

Advanced Biochemistry II

- Independent Recitations with self-prepared lectures
- Grading the exams and assignments
- Helping professors with day-to-day tasks

Lead Trainer, American Society of Plant Biologists, Milwaukee, WI

June 2025

Advanced RNA-seq analysis - beyond differential expression

- Led an online RNA-seq workshop for ASPB, focused on early career plant scientists
- Introduced ways to extract more data from currently existing RNA-seq data
- Talked about SNP, WGCNA, Splice variants, contrast analysis and helixer

Lead Trainer, American Society of Plant Biologists, Honolulu, HI

June 2024

RNA-seq Tools: What You Need, How to Choose, and Basic Skills to Make it Work

- Led an online RNA-seq workshop for ASPB, focused on early career plant scientists

- Talked about basics of RNA-seq experimental design and tools to use
- Information about techniques such as GO, KEGG, WGCNA, impulse models

Mentor, Michigan State University, East Lansing, MI

May 2023 - July 2023

Research Experience for Undergraduates in Plant Genomics

- Guided a visiting undergraduate student on transcriptomic study on intersection of herbicide and drought stress
- Helped gain basics on command line, R scripts, data presentation, literature review
- Identified two Aldo-keto reductase genes that might cause glyphosate resistance in *Conyza canadensis*

Trainer, Michigan State University, East Lansing, MI

October 2022

Ecotek Lab Youth Scientists Visit

- Guided high school students on basics of lab techniques
- Helped run agar plate assays on allelopathic compounds
- Taught sterile techniques for plating on nutrient-rich media

Mentor, Michigan State University, East Lansing, MI

May 2022 - July 2022

Research Experience for Undergraduates in Plant Genomics

- Guided an undergraduate student on RNA-seq analysis of resistant and susceptible *Poa annua* populations
- Molecular docking simulations on Indaziflam and suspected target genes were also conducted

Awards & Achievements

Travel award, American Society of Plant Biology	2025
Agricultural Genome to Phenome Initiative Travel Award, United States, Department of Agriculture and Iowa State University	2024
Travel award, American Society of Plant Biology	2024
Early Career Scientist Travel award, American Society of Plant Biology and, Early Career Plant Scientists	2023
Travel award, College of Graduate School, Michigan State University	2023
Travel award, International Weed Genomics Consortium	2023
Travel award, Plants, Soil Science and Microbes department, Michigan State University	2022
Dudley & Angela Smith Graduate Education Enhancement Fund, Michigan State University	2022
Dr. Gurdev Singh Khush Foundation for Advancement of Agricultural Sciences scholarship, Punjab Agricultural University	2019, 2021
University Merit Scholarship, Punjab Agricultural University	2019